DANIEL JIANG

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EDUCATION

Stanford University, Stanford, CA — **Sep 2024 – June 2026**
M.S. in Electrical Engineering | Concentration in Software & Hardware Systems — **GPA: 3.93/4.00**
University of Michigan, Ann Arbor, MI — **Aug 2020 – May 2024**
B.S.E. in Computer Engineering | Minors in Mathematics and Music — ***summa cum laude***

Coursework: Convex Optimization, Reinforcement Learning, Machine Learning, Parallel Computing, Safety Critical Systems, Linear Dynamical Systems, Computer Vision, Operating Systems, Computer Networks, Data Structures & Algorithms in C++

PROFESSIONAL EXPERIENCE

Waymo (Google Self-Driving Car Project), Mountain View, CA — **June 2025 – Sep 2025**
Software Engineer Intern | Planner Reasoning Team

- Engineered a C++ uncertainty-aware spatiotemporal motion planner for navigating freeway exits without offline maps, leading to 17% fewer hard braking and 5% fewer unnecessary lateral deviations in map mutation scenarios.
- Architected a statistical analysis pipeline that identifies high-entropy uncertainty heatmaps by correlating them with anomalous driving behavior, enabling an automated feedback loop for targeted retraining of the perception model.

NVIDIA, Santa Clara, CA — **May 2024 – Aug 2024**
Software Engineer Intern | Autonomous Vehicles Ground Truth Perception Team

- Developed a 3D multi-view stereo reconstruction pipeline in Python using COLMAP, SimpleRecon, and visual-inertial odometry—enabling accurate temporal ground truth generation on LiDAR-less consumer fleet data.
- Integrated and deployed an End-to-End low-latency data factory pipeline for closed-loop perception ML training.

Aurora, Mountain View, CA — **May 2023 – Aug 2023**
Software Engineer Intern | Autonomous Driving Onboard Localization Team

- Built a C++ lane-detection verification system to identify erroneous EKF inputs and support lane-level alignment through geometric analysis with Bezier interpolation; deployed on all vehicles with 97% precision and 80% recall.
- Designed a low-latency LiDAR instance segmentation C++ library using a novel momentum-based DBSCAN point cloud clustering algorithm; deployed onboard for 3D lane detection segmentation.

Robert Bosch LLC, Plymouth, MI — **May 2022 – Aug 2022**
Software Engineer Intern | Advanced Driver Assistance Systems Team

- Adapted an algorithm to detect co-channel FMCW radar interference and characterize sections of detection data by radar velocity, validating the interference pattern of Bosch's parking radar sensors.

RESEARCH EXPERIENCE

Stanford Robotics and Embodied Artificial Intelligence Lab (REAL), Stanford, CA — **Dec 2024 – Present**
Research Assistant | Advisors: Prof. Shuran Song & *Prof. Karen Liu*

- Developing ToddlerBot, an open-source humanoid platform for loco-manipulation reinforcement learning, to enable seamless zero-shot policy transfer from simulation to real-world tasks; paper submitted to ICRA 2026.
- Integrated a stereo depth estimation pipeline with FoundationStereo, achieving 7× faster inference using TensorRT on a Jetson Orin Nano and improving RMSE by 40% with temporal depth gradients for consistency.

University of Michigan Transportation Research Institute, Ann Arbor, MI — **May 2021 – May 2024**
Undergraduate Research Assistant | Advisor: Prof. Matthew Reed

- Published open-source GUI software using Python, Qt, and VTK to generate 3D parametric finite-element body shape models for crash protection studies; awarded 2021 Michigan Student Research Competition 3rd place honor.
- Devised a specialized ICP alignment program and a watertight surface reconstruction method for Kinect 3D scans.

LEADERSHIP EXPERIENCE

Michigan Mars Rover Team, Ann Arbor, MI — **October 2020 – May 2024**
Simulation Team Lead & Senior Technical Advisor

- Placed 1st in both the University Rover Challenge (out of 99 universities) and the Canadian International Rover Challenge in 2022; led a team of 20 engineers in developing the Navigation Simulation System (demo).
- Developed an Unreal Engine C++ 3D simulator and a TypeScript trajectory simulator to replicate rover's behavior on Mars' terrain, simulate IMU Gaussian noise, and enable remote algorithm testing during the pandemic.

SKILLS

Programming: C++, Python, Bash, SQL, Verilog, JavaScript, MATLAB, TensorRT, ONNX, OpenGL, OpenCV, PyTorch, Qt
Systems: LiDAR, IMU, Radar, AWS, GCP, Jetson, ROS, Gazebo, Eigen, Bazel, Protobuf, VTK, Docker, Gerrit, Mercurial, Git
Away from keyboard: Basketball analytics, Hamilton, Skiing, Poker, Budae jjigae after exams, Venetian seafood risotto